

Popular Simulation Platforms for the Internet of Vehicles

In these days of traffic congestion and autonomous driving, software that connects pedestrians and vehicles with governing bodies is the need of the hour. Open source simulation platforms for the Internet of Vehicles are enabling just that.

The Internet of Vehicles (IoV), a sub-segment of the Internet of Things (IoT), enables smart vehicular networks, smart urban infrastructure, wireless integrated smart ambulances, and more. Smart vehicular networks are today used for traffic law enforcement, as a part of e-governance and smart citizen services. IoV helps vehicles to communicate with each other with respect to the status of the traffic, keeping roads clear and safe.

This wireless technology is particularly useful for autonomous driving.

Smart cities are about keeping us safe, helping our planet, and making life a lot more fun. Vehicles in smart cities and smart vehicular networks send information to each other using special communication called vehicle-to-everything (V2X). V2X helps vehicles communicate with each other, sends signals to traffic lights, and even provides information to people walking around.

It helps car drivers be aware of what's happening around them. It can make driving safer by giving a warning of any danger ahead and can help ambulances get through traffic really fast.

Simulation tools and scenarios are being used for research and development in the fields of smart cities and vehicular networks. These provide a controlled, cost-effective environment to model, analyse, and optimise complex urban transportation systems before real world implementation. They analyse complex scenarios to mitigate risks and ensure roads are safe for everyone.

A few important free and open source platforms for IoV simulation for research are described briefly here.

SUMO (Simulation of Urban Mobility)

URL: <https://eclipse.dev/sumo/>

SUMO is powerful and multi-functional open source software supporting microscopic traffic simulation. It is used for the simulation of IoV-based scenarios in smart cities. It can simulate smart traffic forecasting, evaluation of traffic lights, automated driving, vehicle communication, and urban mobility planning and analysis. It combines with artificial intelligence and data science for multi-dimensional data evaluation and predictions.

Eclipse SUMO supports automated driving and vehicle communication simulations so that smart IoV scenarios can be investigated effectively with the use of emerging technologies. Intermodal traffic systems including road vehicles, public transport, and



Figure 1: Smart city and wireless connected vehicles